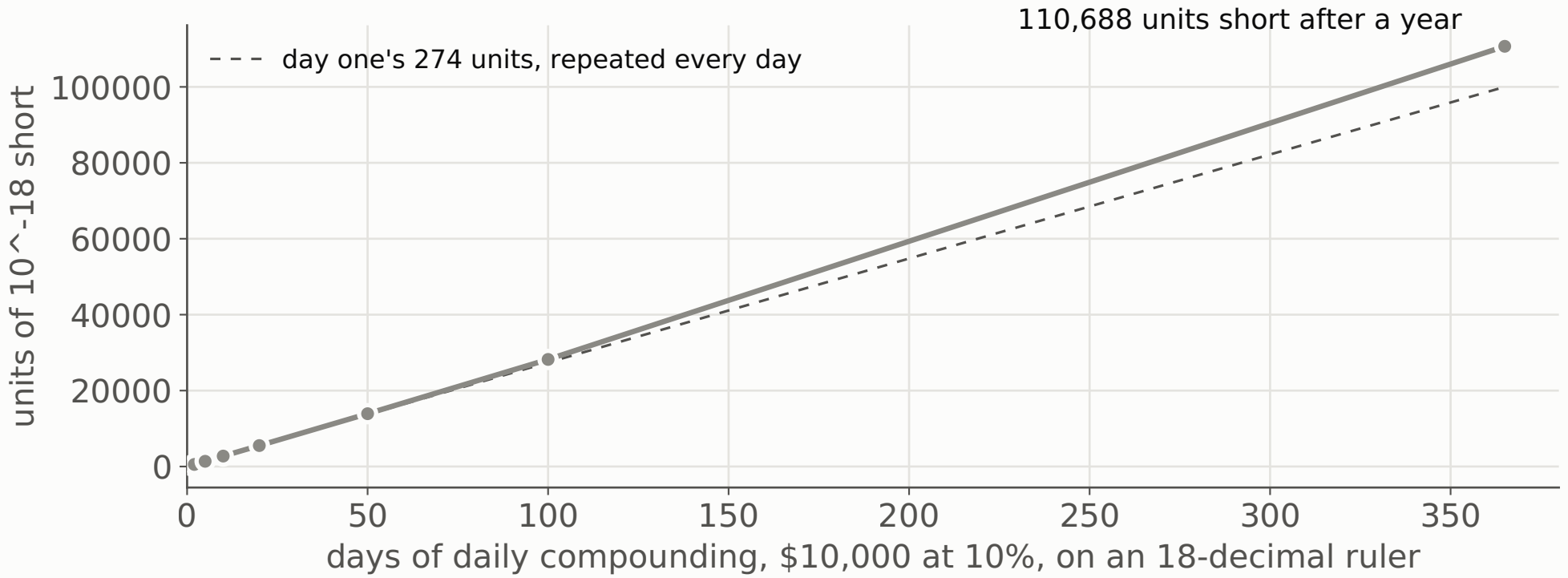
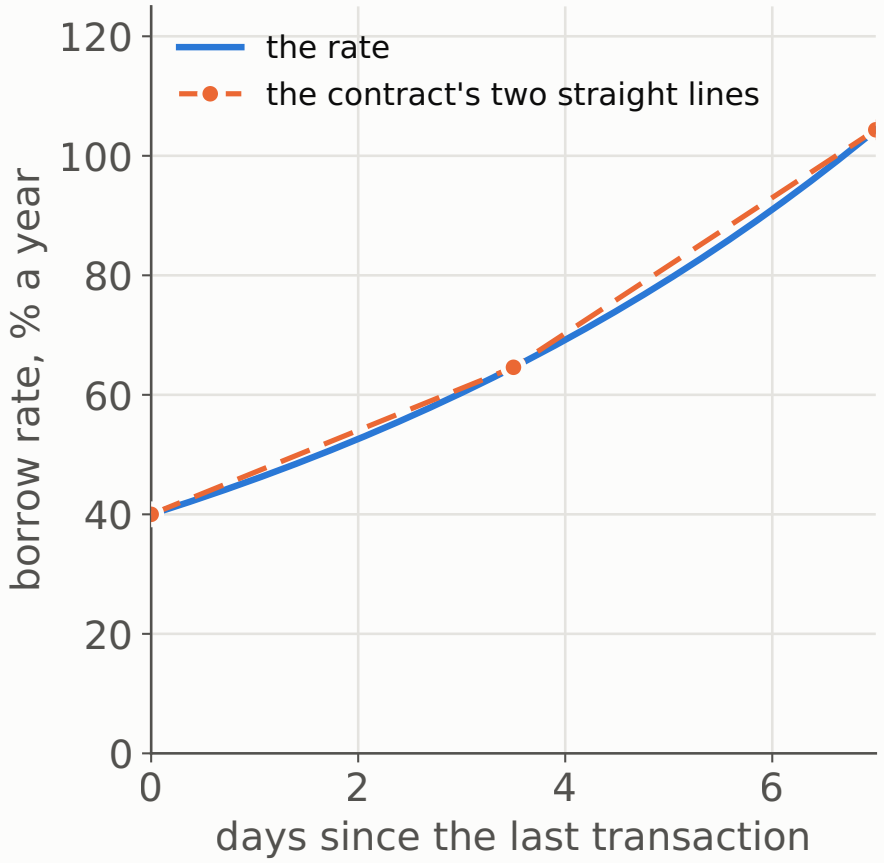
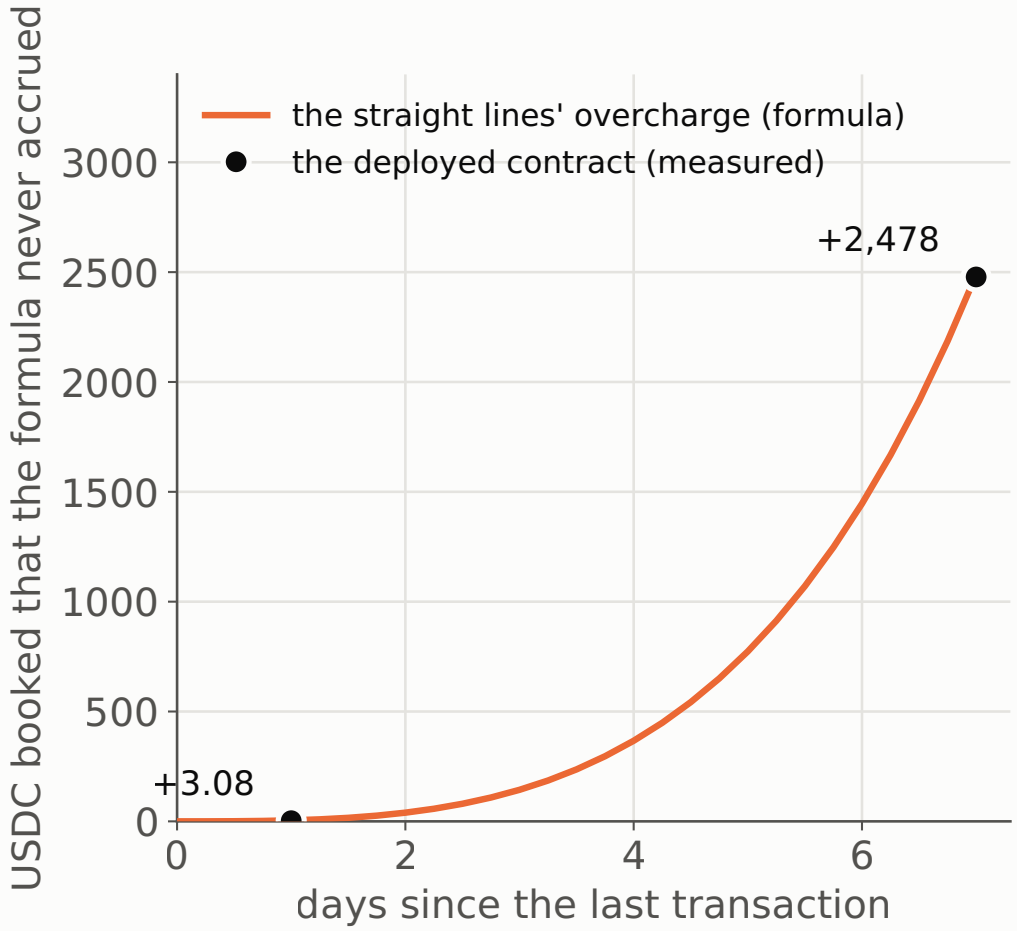
Dust is Over



$10,000 at 10%, compounded daily on an 18-decimal ruler. docs/benchmarks/ladder.json: compound

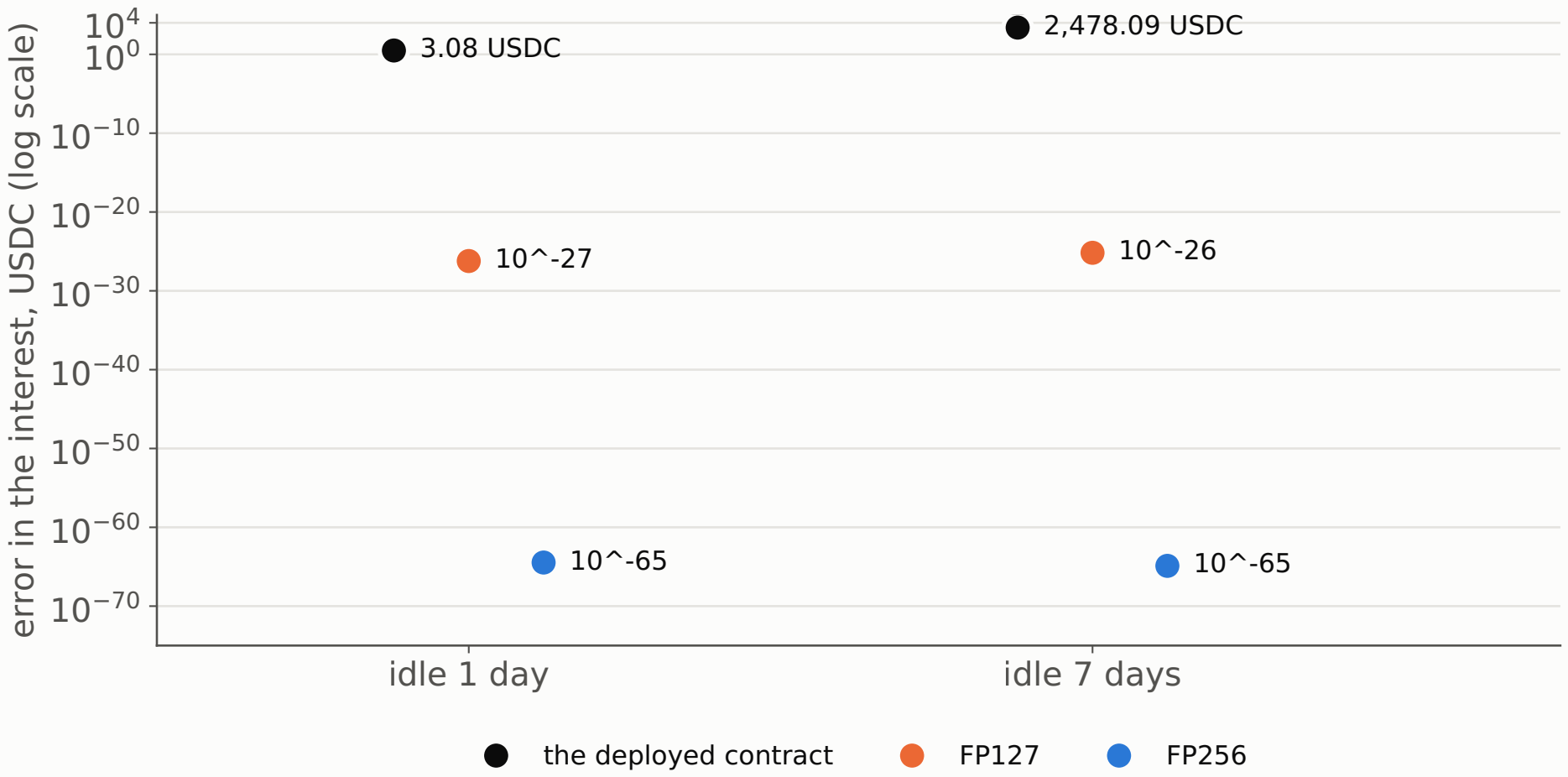
Ten million dollars, untouched for a week





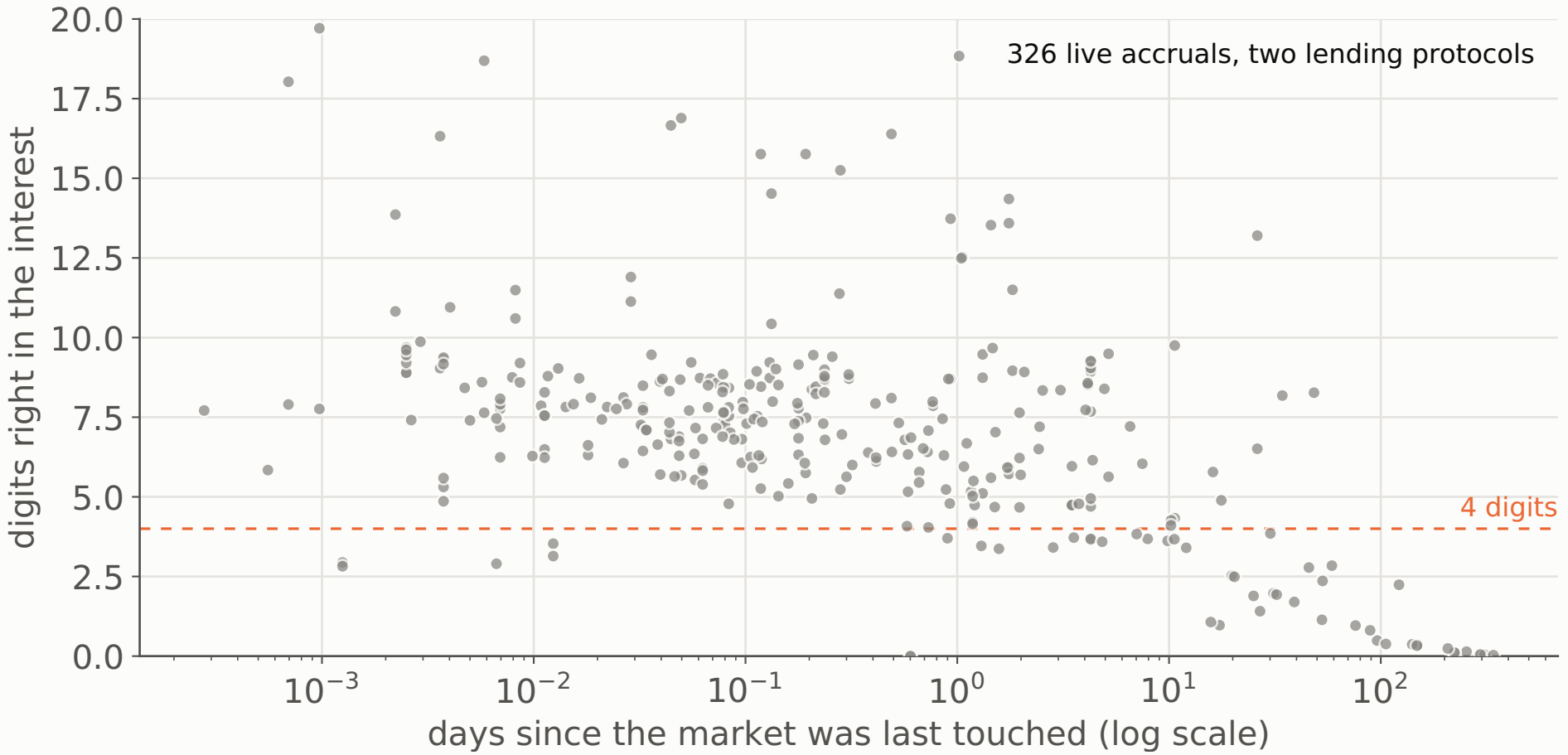
10M USDC borrowed, utilization 100%, target rate 10%. The rate model and market as deployed (byte-equal to mainnet) vs the stated formula at 200 digits. docs/benchmarks/accrual-deployed.md

The same calculation, three ways



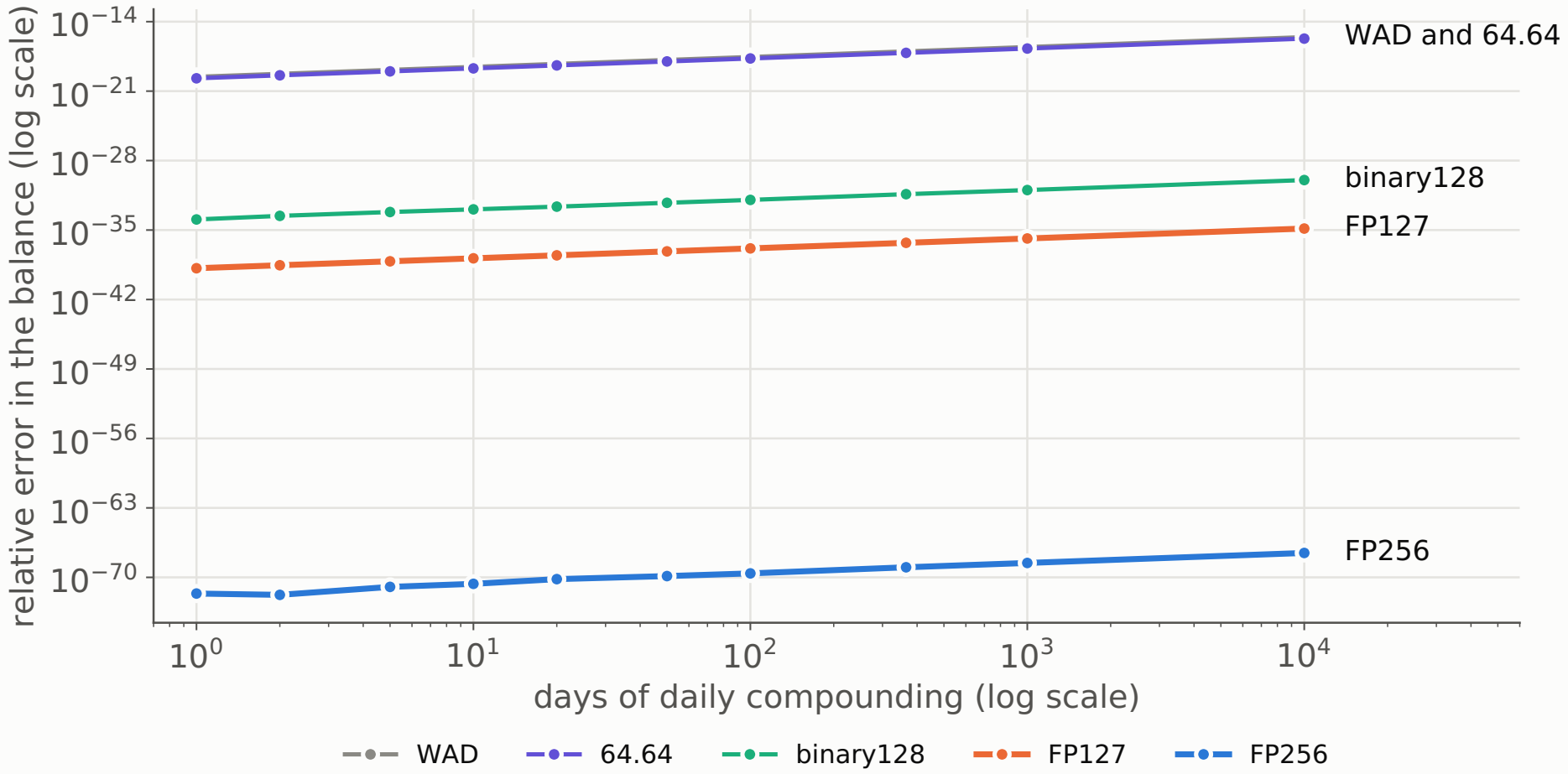
10M USDC borrowed, utilization 100%, target rate 10%. The rate model and market as deployed (byte-equal to mainnet) vs the stated formula at 200 digits. docs/benchmarks/accrual-deployed.md

It happens in live markets



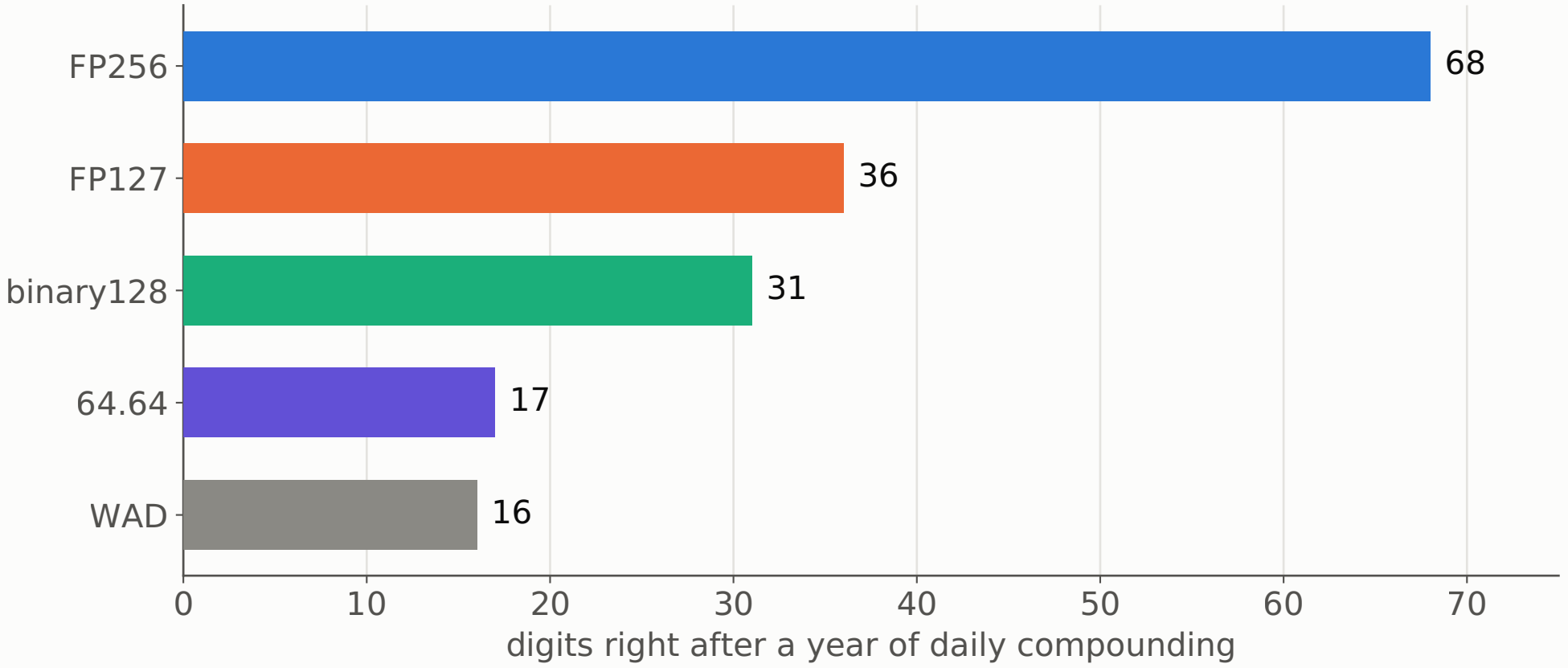
Every live lending reserve and market with $10k+ borrowed, block 26,206,698, accrued by the deployed code. docs/benchmarks/accrual-deployed.md

Rounding with nowhere to go



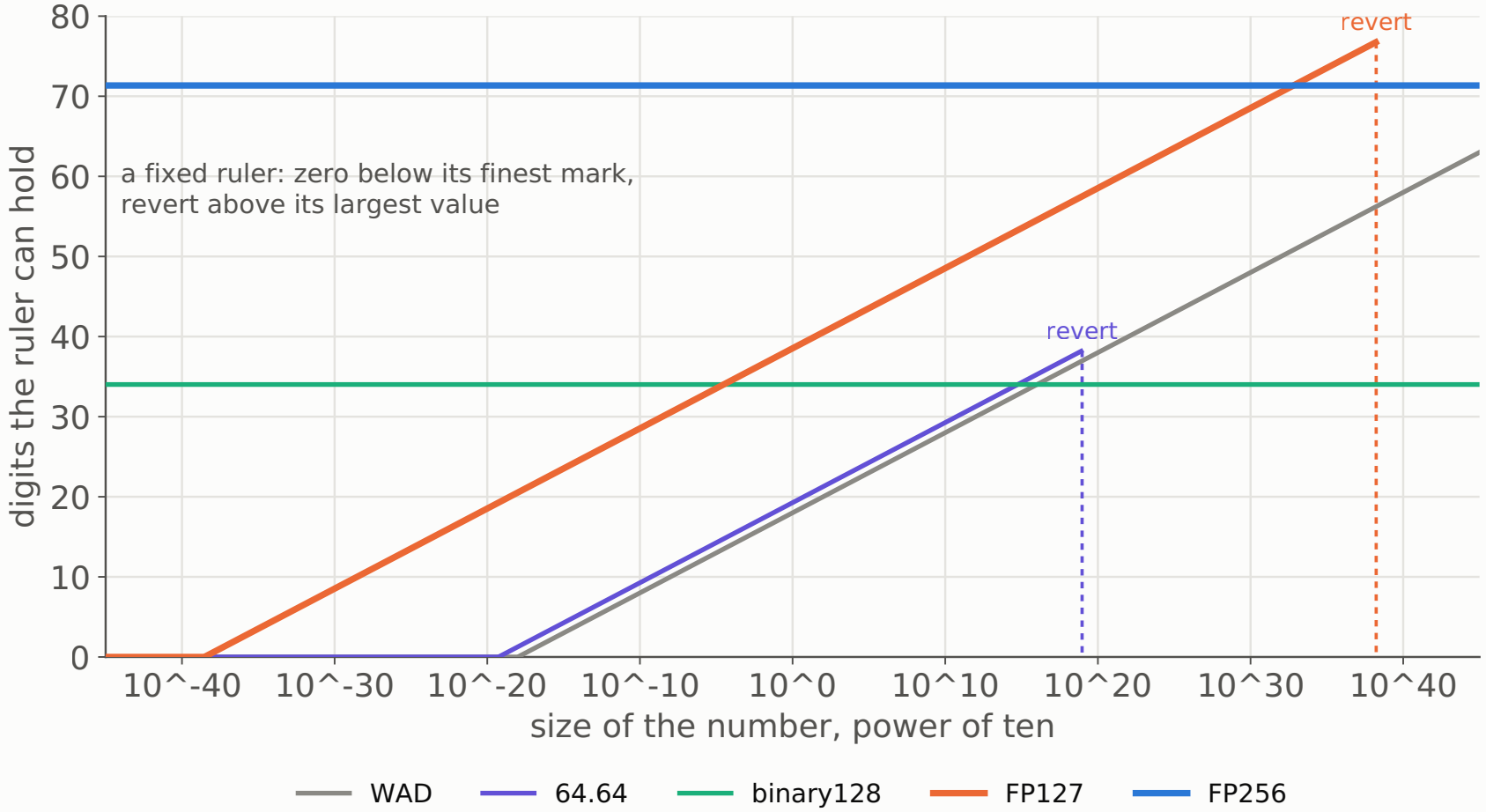
Daily compounding, reference input. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

Goal one: round once



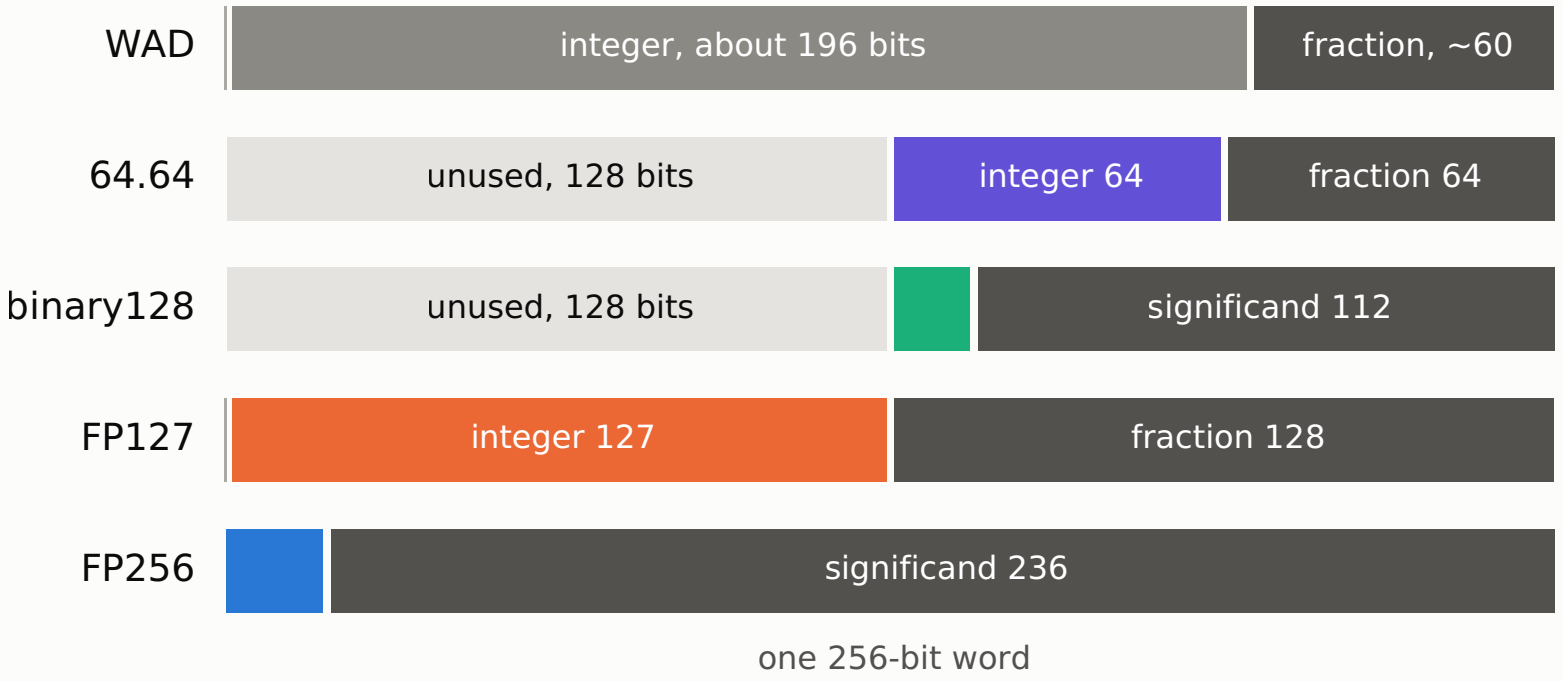
A year of daily compounding, $10,000 at 10%. docs/benchmarks/ladder.json. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

Goal two: never run out of ruler



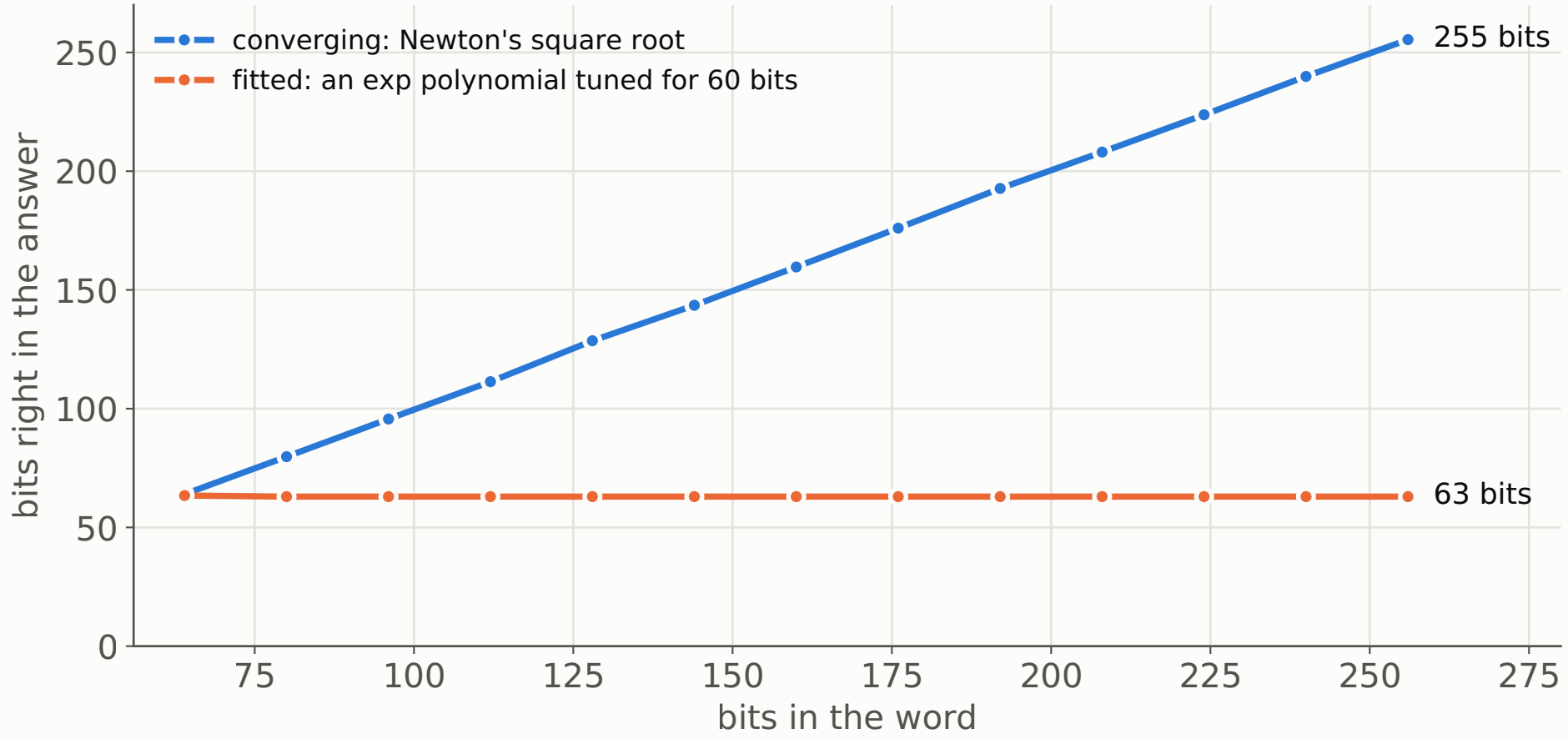
Exact properties of each format: the finest mark, the largest value, the significand.

Five ways to spend one word



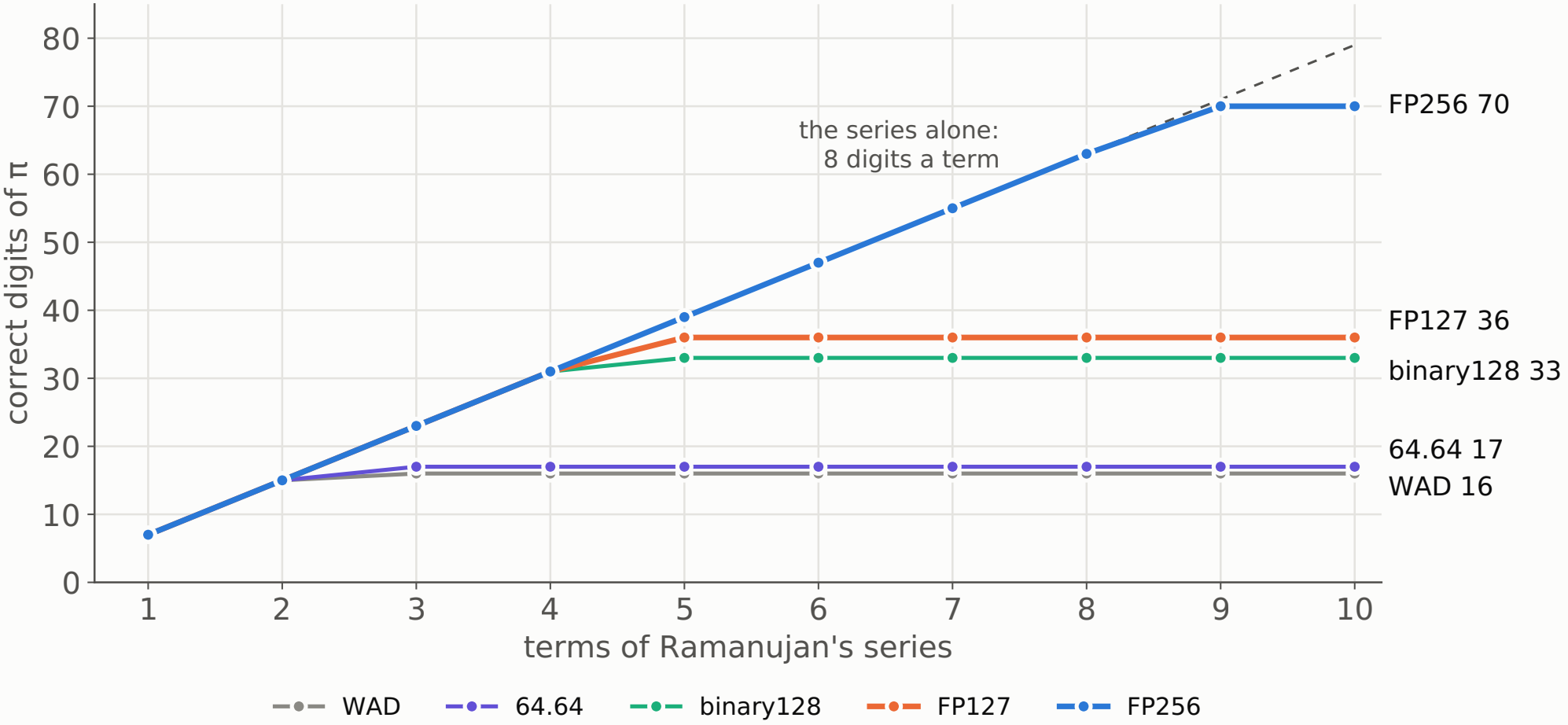
Bit layouts of each format in a 256-bit word.

Converge, or fit



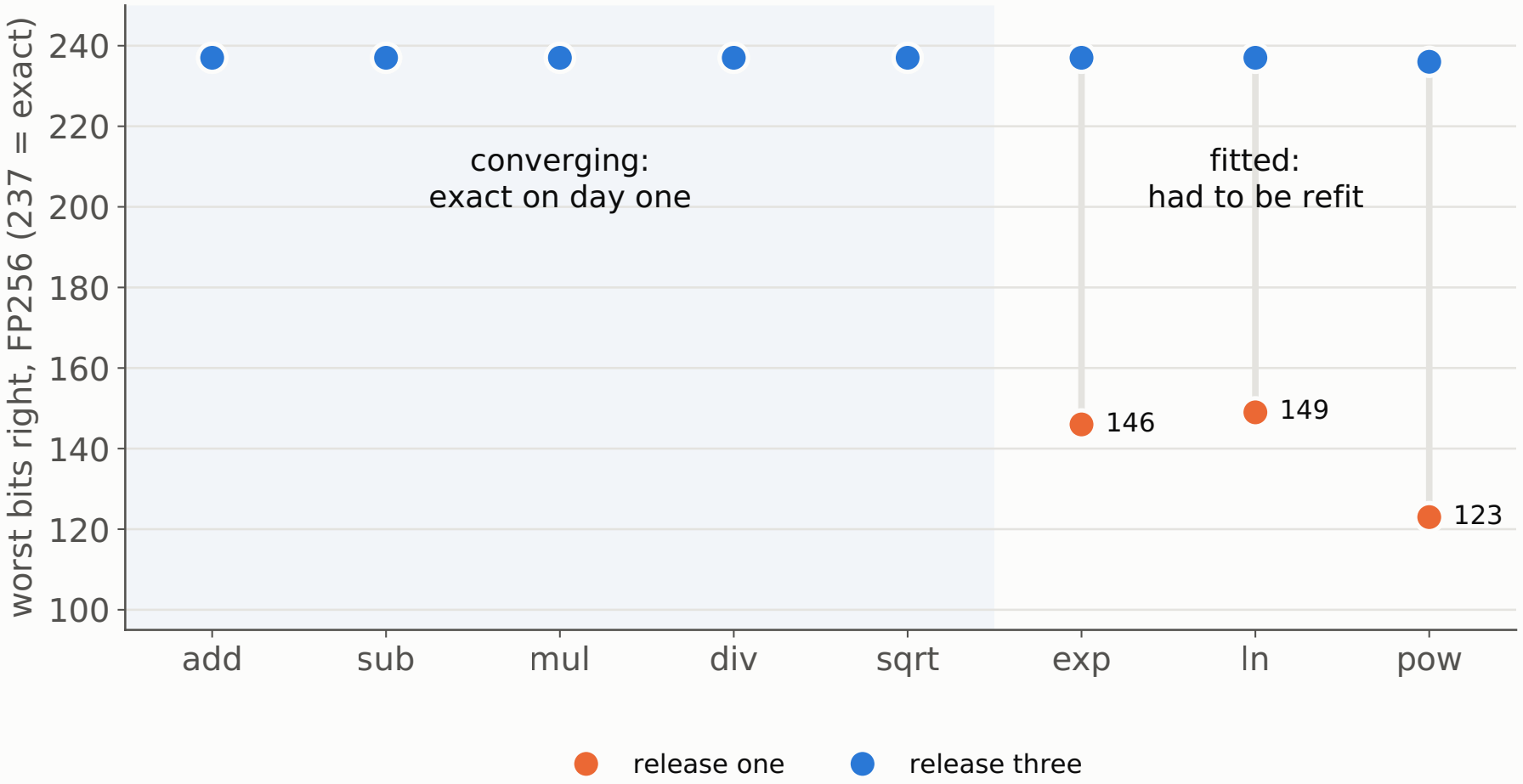
Computed in exact arithmetic at each width: Newton's sqrt(2), and the shortest Taylor polynomial for exp within 2^-60 on [0, ln 2], at x = ln 2.

The series never stops. The ruler does.



Ramanujan's series through each library's own sqrt, mul and div. docs/benchmarks/pi.md

FP256 proved it by accident



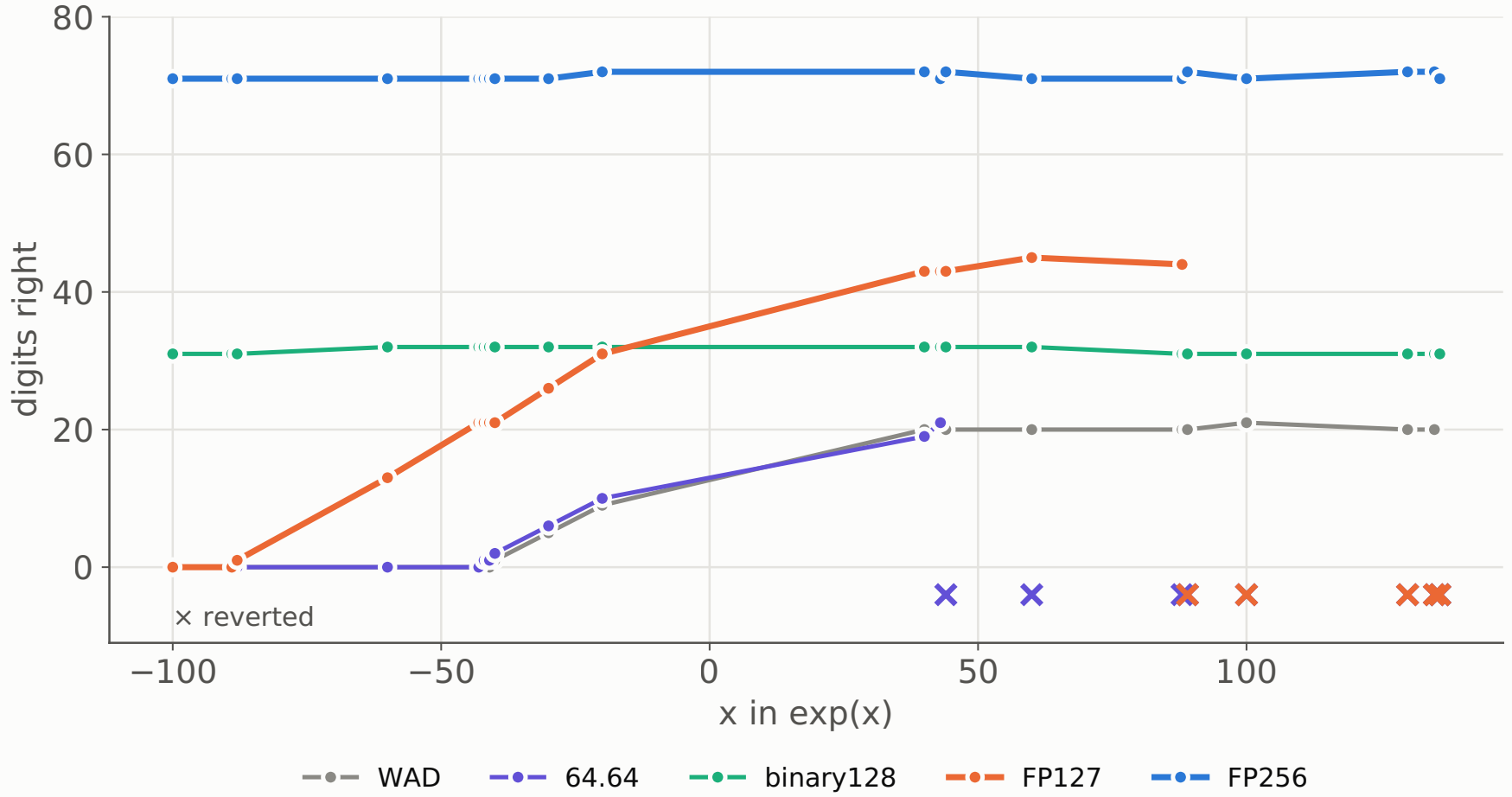
Worst bits over the committed vectors. Release one: contracts/deployments/FP256.md; release three: docs/fp256/precision.json.

The walls

	WAD	64.64	binary128	FP127	FP256
ln(1+1e-9)	9	10	25	30	62
log1p(1e-30)	zero	zero	4	8	72
exp(-40)	1	2	32	21	71
exp(-42)	zero	1	32	21	71
exp(-89)	zero	zero	31	zero	71
exp(89)	20	revert	31	revert	72
mul(1e-10,1e-10)	zero	zero	34	18	71
div(1e-9,1e12)	zero	zero	33	17	71
sqrt(1e-30)	zero	zero	34	8	71
pow(1.0001,-887272)	zero	zero	29	4	65
pow(1.0001,887272)	12	revert	29	revert	65
normCdf(-9)	zero	1	31	21	71
lambertWm1(-1e-10)	n/a	n/a	n/a	30	71

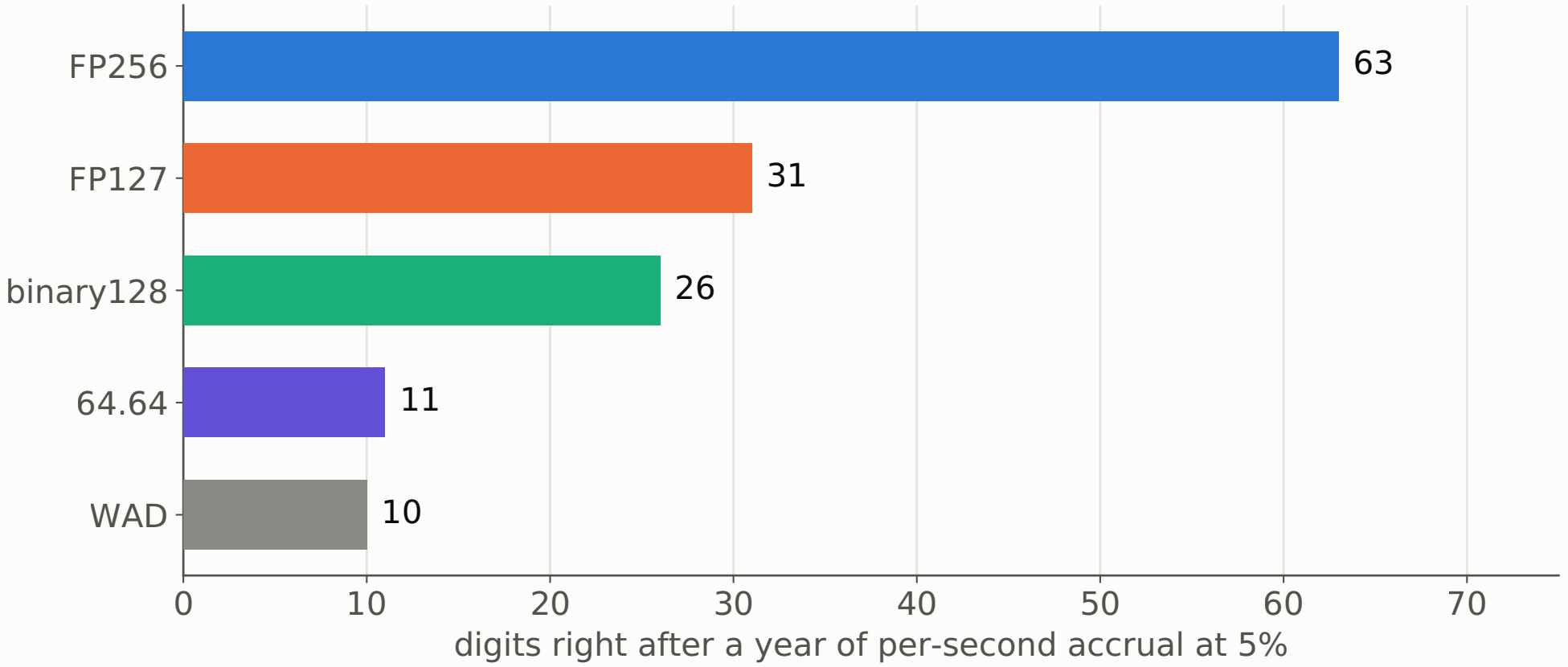
One call each, at the end of the domain, against 200 digits. docs/benchmarks/edges.json. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

exp, end to end



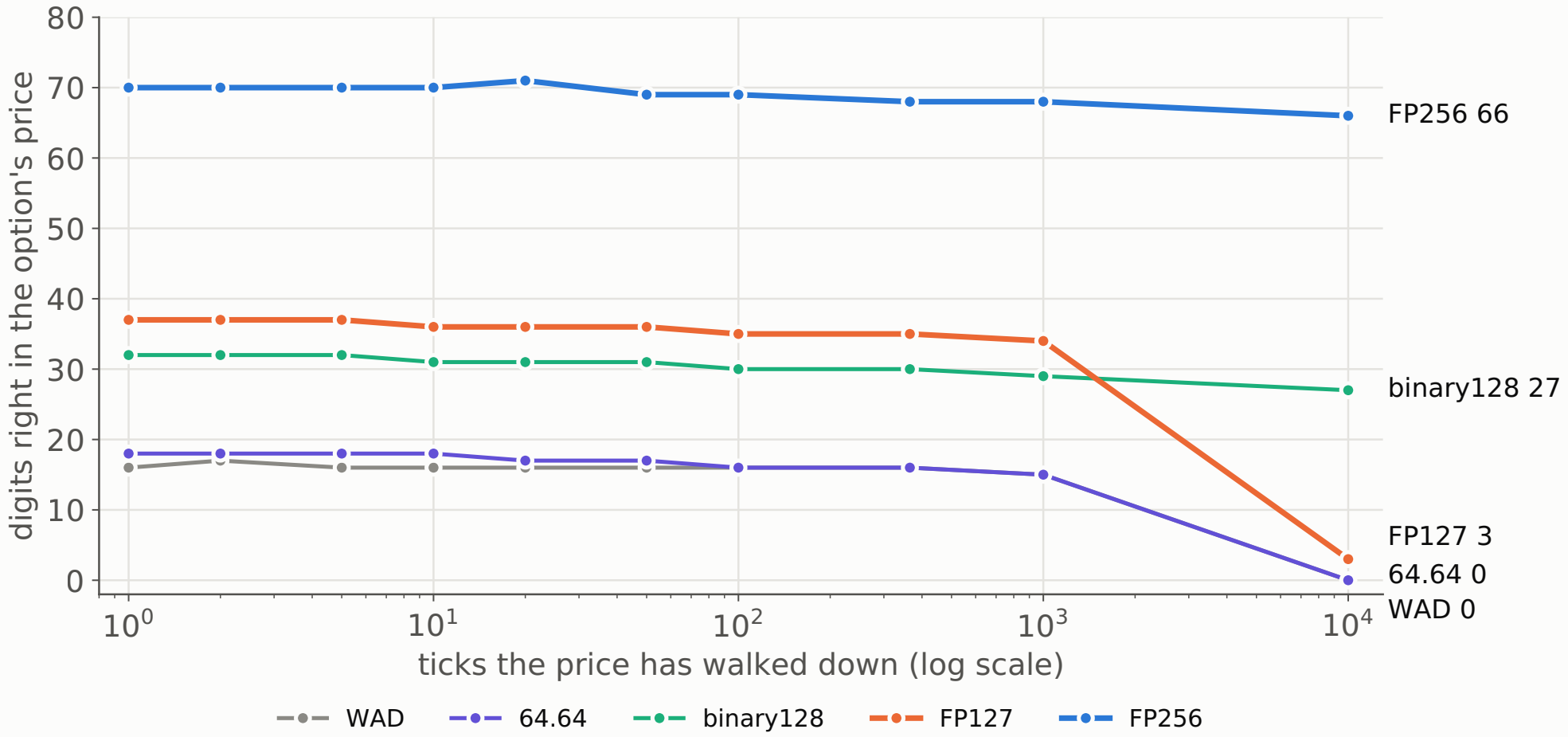
docs/benchmarks/edges.json: exp probes. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

A rate per second



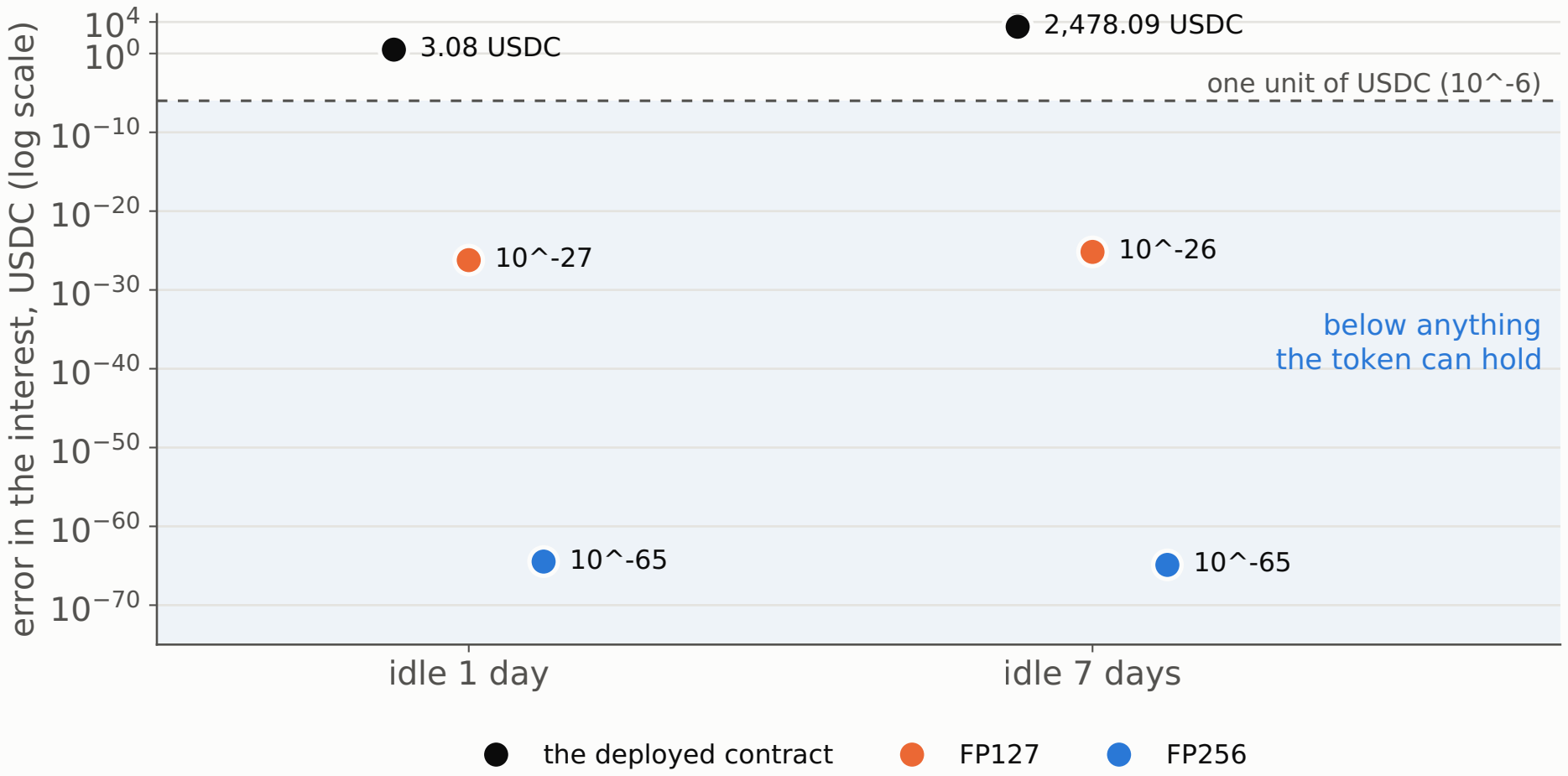
5% a year accrued per second for a year. docs/benchmarks/ladder.json: accrual-per-second. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

An option, walked down



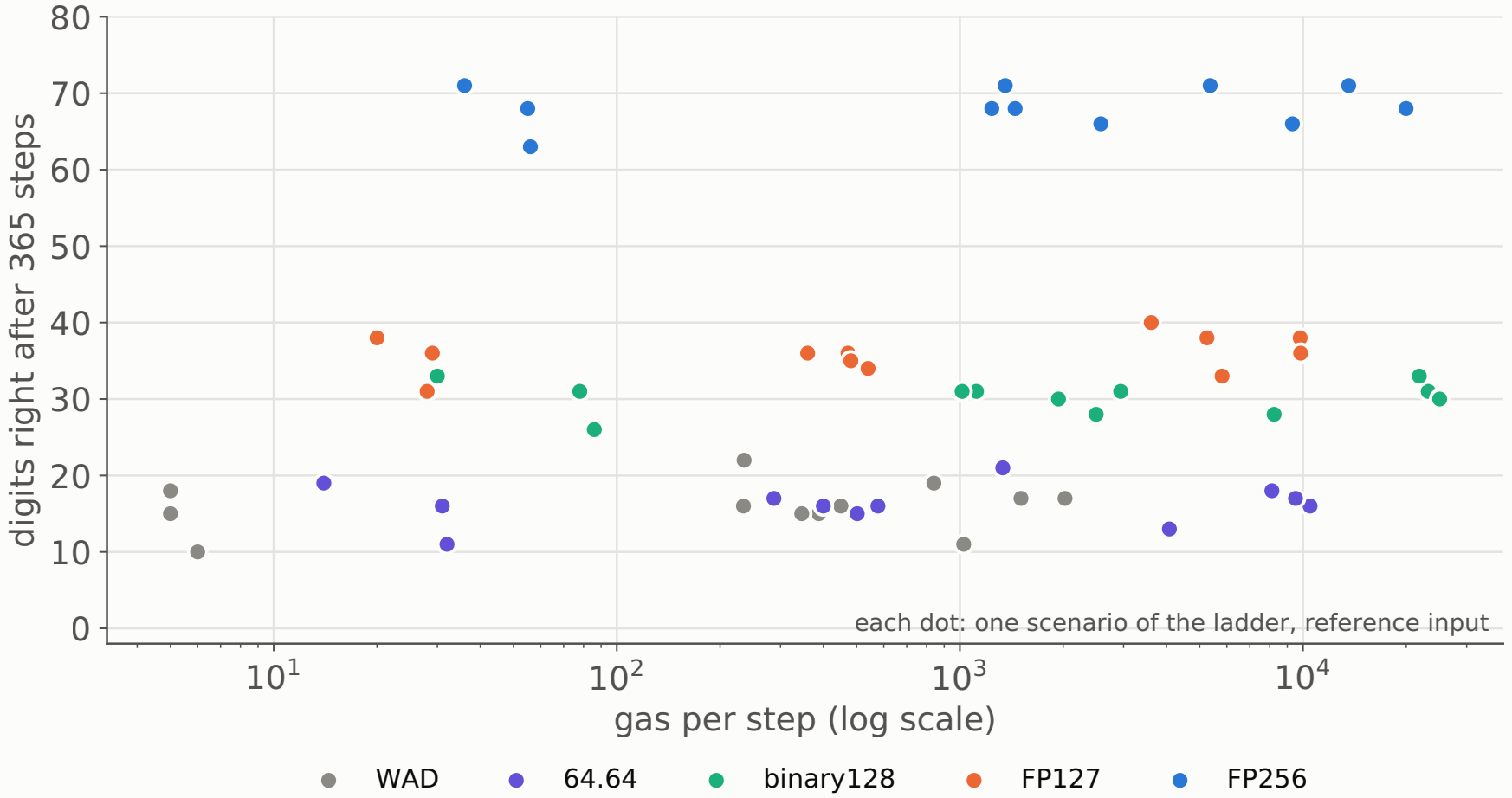
Normal-distribution option priced each tick as the price walks to d1 = -12. docs/benchmarks/ladder.json: deep-tail-option. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

The same market, rounded once



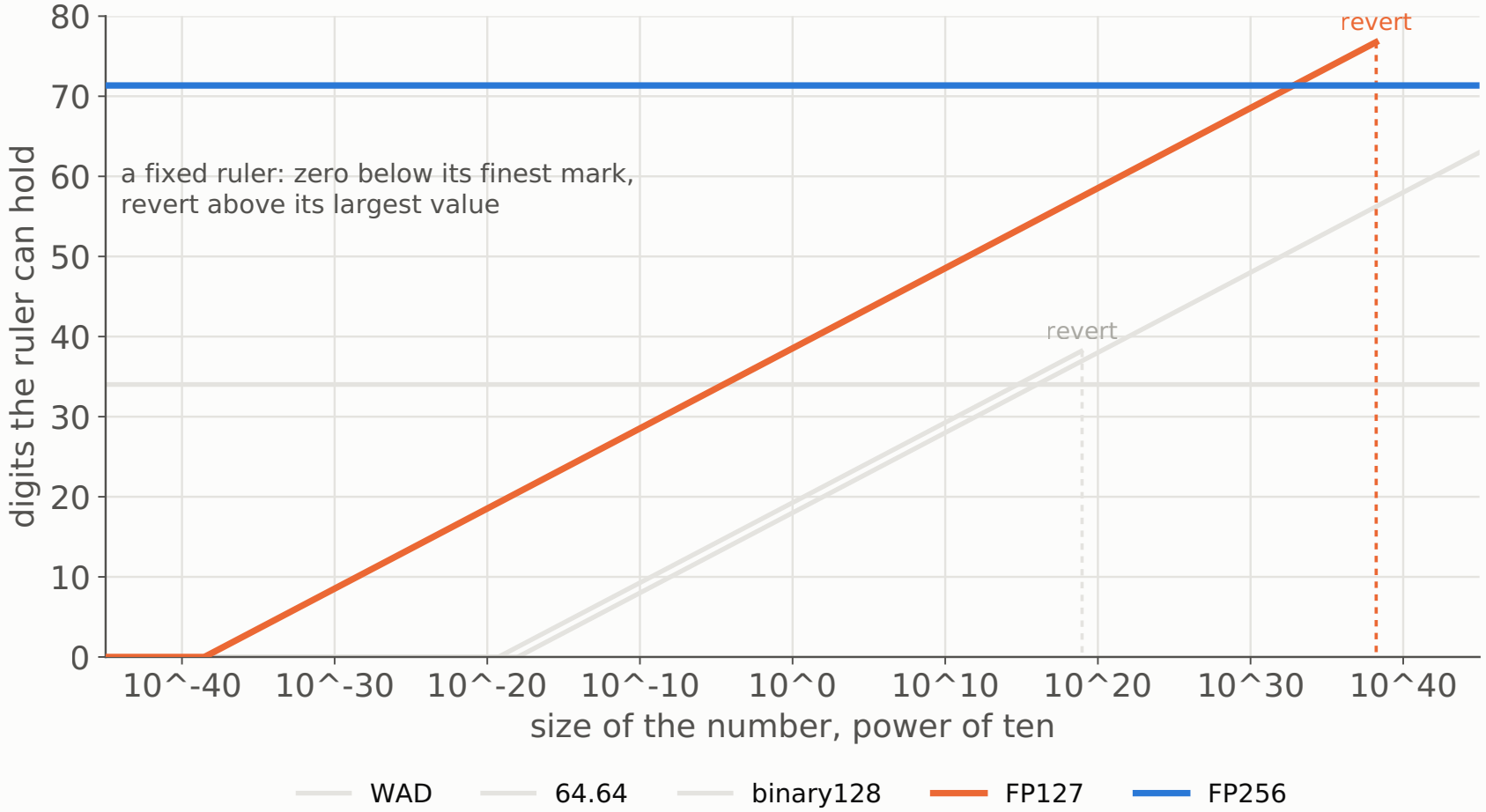
10M USDC borrowed, utilization 100%, target rate 10%. The rate model and market as deployed (byte-equal to mainnet) vs the stated formula at 200 digits. docs/benchmarks/accrual-deployed.md

What the digits cost



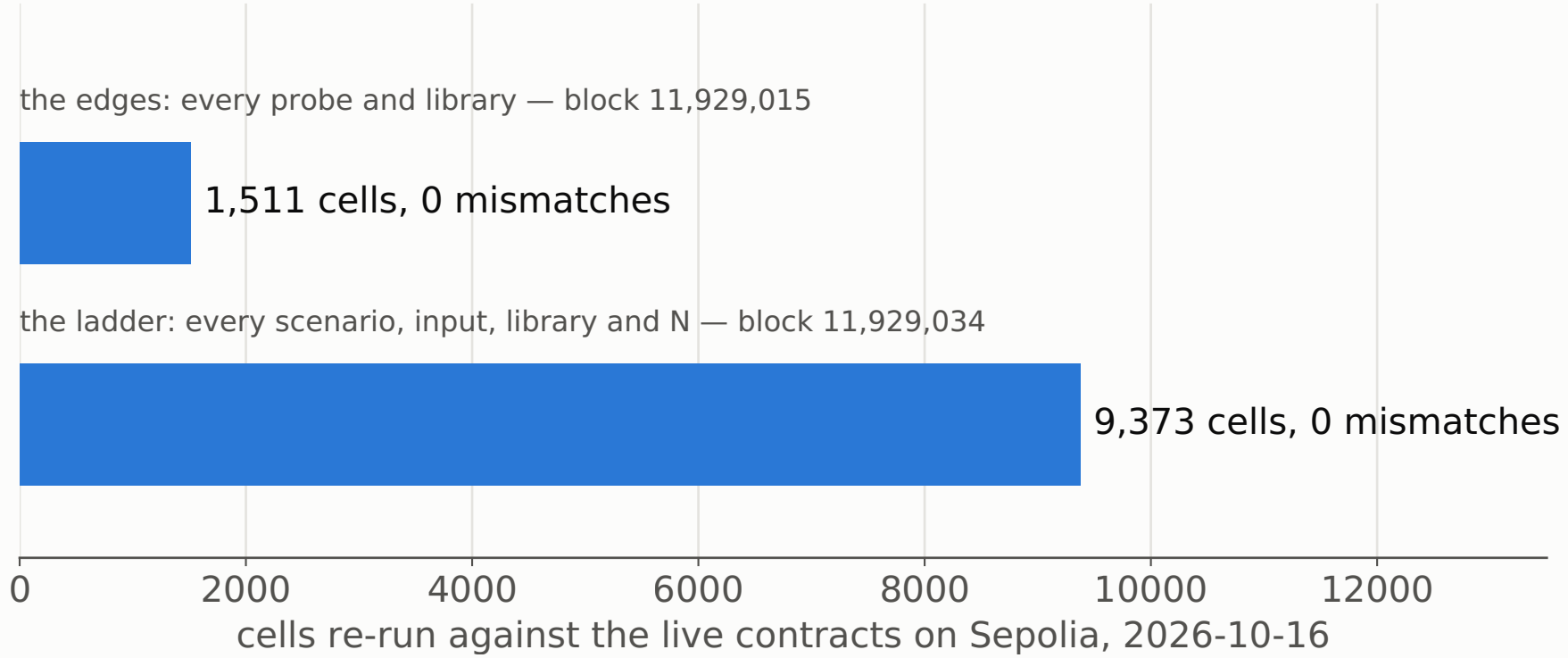
Gas per step and digits after 365 steps, every ladder scenario. docs/benchmarks/ladder.json. WAD = an 18-decimal library; 64.64 and binary128 = the two ABDK formats.

Both goals, met



Exact properties of each format.

Checked on chain



contracts/deployments/RELEASE.json

Use FP127 and FP256

FP127

`0x6cFbCB928Ac068E84Ef793Dd1DFce4f8e9Cac1d1`

FP256

`0x67eFfb11bF93c79a2B12cc7872cBaC12b3eB61be`

blockulator.com · github.com/securedataresearch/blockulator